

Expansion on Exploring Common Trends in Online Educational Experiments

Aaron Haim (ahaim@wpi.edu) Worcester Polytechnic Institute, 100 Institute Road, Worcester, Massachusetts, United States

Ethan Prihar (ebprihar@wpi.edu) Worcester Polytechnic Institute, 100 Institute Road, Worcester, Massachusetts, United States

Stacy T. Shaw (sshaw@wpi.edu) Worcester Polytechnic Institute, 100 Institute Road, Worcester, Massachusetts, United States

Adam Sales (asales@wpi.edu) Worcester Polytechnic Institute, 100 Institute Road, Worcester, Massachusetts, United States

Neil T. Heffernan (nth@wpi.edu) Worcester Polytechnic Institute, 100 Institute Road, Worcester, Massachusetts, United States

Abstract

In recent years, online learning platforms have slowly become more commonplace. As such, there is a growing need to evaluate the effectiveness of content, education, and supplemental tutoring within these platforms. In 2021, the National Center for Education Research (NCER), managed by the Institute of Education Sciences (IES) under the U.S. Department of Education, granted five platforms two million dollars each, and a research network three million dollars to support the field in applying to three platforms, towards this goal to allow researchers to conduct experiments within these platforms, collaborate, and share results (IES, 2021). In addition, the NCER has just announced for 2023 to fund 10 external researchers to propose experiments and secondary data analyses within one of the five funded platforms (Federal Registry, 2022). IES expects to fund ten teams, each with up to 400,000 dollars, to propose and conduct experiments with these platforms. As representatives of ASSISTments (Heffernan & Heffernan, 2014), one of the funded online learning platforms (Heffernan, 2021), we are happy to share with the CODE conference attendees the release of the largest dataset for experiments in education, such that we can provide external researchers the ability to create better policies to personalize learning. CODE has had many presentations on different ways to detect heterogeneous treatment effects (e.g., Susan Athey), so these datasets will allow the CODE community to try their hand to see if they can create and learn personalization policies. For an example of a policy you might be able to learn, a student with higher knowledge should be given Condition A while a student with lower knowledge should be given Condition B. For instance, Rafferty and colleagues (Rafferty, Ying, & Williams, 2019), working with a dataset we released in 2016 (Selent, Patikorn & Heffernan, 2016), reported that they found some personalized learning policies.

These datasets have already won two awards this summer at the *9th ACM Conference on Learning at Scale* and the *15th International Conference on Educational Data Mining* as the best dataset of the conferences, but in this work for CODE, we are planning to expand upon the 50

Experiments-2021 dataset used in the paper “Common Trends in Online Educational Experiments” (Prihar, Ostrow et al., 2022) to better support experiments and secondary data analysis performed by external researchers. Specifically, this work will create the *50 Experiments+2022* dataset, which provides the raw log data for each problem across the fifty experiments and all of the clickstream and action data students performed prior to the experiment period within the platform, adhering to the Standards for Excellence in Education Research (SEER) principles provided by IES for conducting Open Science research (IES, 2022). Answers that students have to provide themselves rather than through preset options will be stripped to avoid revealing personally identifiable information. In addition, we will discuss some of the implications of moving the educational field towards Open Science digital experimentation such that researchers, if willing, preregister their experiments and make their data and analysis publicly available through the Open Science Foundation (OSF). The expanded dataset along with an automated setup process can be found in an [OSF project](#)¹.

Within **open textbooks**, which provide problems that are free and publicly accessible, researchers can propose experiments at scale which compare on-demand assistance commonly in the form of hints and explanations, known as **student-supports** within ASSISTments, for any given problem through the Special Content System, formerly known as TeacherASSIST. Using the Special Content System, we found that when tutoring is requested by a student, providing a *student-support* compared to immediately giving the answer caused more student learning (Patikorn et al., 2020). Evidence was additionally reported between any two authors on which one was more effective at improving student learning based on their first request of a *student-support* by that author (Prihar et al., 2021). We have also published our own attempt on conducting experiments for personalizing learning for students through *student-supports* (Prihar, Haim et al., 2022; Automatic Interpretable Personalized Learning). This dataset won the Best Dataset award at the *9th ACM Conference on Learning at Scale* (Prihar, Haim et al., 2022; Automatic Interpretable Personalized Learning Datasets).

Within **skillbuilders**, assignments which can be completed once the student completes three problems correctly in a row, researchers can propose more finely tuned experiments at scale for a given problem set through the Ed-Tech Research Infrastructure to Advance Learning Science (E-TRIALS) within ASSISTments (Krichevsky et al., 2020). Prihar reported on and made publicly available a large dataset containing fifty experiments, containing 50,752 instances across 30,408 students, run by researchers in E-TRIALS since February, 2019 (Prihar, Syed et al., 2022). It received the Best Publicly Available Educational Data Set award at the *15th International Conference on Educational Data Mining* (Prihar, Ostrow et al., 2022). Some of the experiments within the dataset included fourteen experiments across 12,243 students injecting motivational messages and videos into the assignment, five experiments across 2,492 students comparing tutoring provided in a text-based versus a video format, and two experiments across 4,057 students comparing filling in the correct answer versus selecting from multiple preset options for a given problem. The experiments found that motivational messages had a negative impact on student learning, students learned more from tutoring in a video format compared to

¹ <https://doi.org/10.17605/osf.io/m2jqe>

text-based, and students learned more where they filled in the correct answers compared to selecting the answer from multiple options.

This work releases the expansion publicly to supplement **Open Science** practices. Open science has gained traction since the early 2010s (Spellman, 2015), describing and making the methods, data, materials, and results of research public and accessible to all (Vicente-Saez, 2018). In addition, open science practices are meant to improve the thoroughness of explanations, reproducibility during review of the work, and replicability for confirming the outcomes. In the mid 2010s, a number of studies alerted numerous disciplines to the lack of reproducibility and replicability of their research (Baker, 2016). In response, the observed disciplines, along with some others, more actively adopted open science principles, improving rigor and clarity in many subsequent works.

One of the topics within open science which is relevant to this dataset is **Open Data**. Open data is a term in which a dataset or datasets is made publicly accessible and can be used without restriction (Murray-Rust, 2008). To declare open data, a dataset is either given a permissive license (e.g., Creative Commons Attribution 4.0 International² (CC-BY-4.0) or Creative Commons Attribution Share Alike 4.0 International³ (CC-BY-SA-4.0)) or released into the public domain (e.g., Creative Commons Zero v1.0 Universal⁴ (CC0-1.0) or Open Data Commons Public Domain Dedication and License (PDDL) v1.0⁵ (PDDL-1.0)).

Sometimes datasets cannot be made openly available: a dataset may contain non-anonymized personally identifiable information (PII), the dataset (such as ours) has some restrictions that prevent full open access (non-commercial, usage restrictions), etc. In these cases, we can attempt to comply with as many open data practices as possible. Datasets with non-anonymized PII can be uploaded to a protected access data repository⁶ with the necessary steps to obtain the data, or the dataset can be given a list of steps and requirements to request access from the author of the work. This work contains four clauses that prevent full open access but still allows us to make the dataset publicly available: a non-commercial clause, as we want our dataset to be used for improving freeware tools and concepts; a no derivatives clause, as we want people to always use the original dataset and to not make unsubstantiated claims through poorly designed research; a full transparency clause, to promote open science practices and force users to provide evidence to their claims when using our dataset; and a deanonimization restriction clause, to prevent users from risking or harming people within the ASSISTments platform. However, we do provide exemptions from these clauses with an associated licensing agreement with the ASSISTments Foundation.

Within the discipline of education technology, the adoption of open science practices is sparse or stagnant at times (Nosek, 2022). There are numerous reasons for this: researchers are not

² <https://creativecommons.org/licenses/by/4.0/>

³ <https://creativecommons.org/licenses/by-sa/4.0/>

⁴ <https://creativecommons.org/publicdomain/zero/1.0/>

⁵ <https://opendatacommons.org/licenses/pddl/1-0/>

⁶ <https://osf.io/tvyxz/wiki/8.%20Approved%20Protected%20Access%20Repositories/>

well informed about the literature behind making their work openly available, concerned about whether they can make their work openly available, inexperienced in making their work openly available causing difficulty or impossible circumstances when reproducing or replicating, etc. As such, resources and education on open science practices need to be made more readily available to provide researchers with the necessary understanding to make their work more openly accessible—whether through methodology, dataset, analysis, preregistration, etc.—and to mitigate any potential issues when future researchers choose to reproduce or replicate existing work.

This work provides some starting work and examples for complying and requiring other researchers to comply with open science practices. While the dataset is not fully open access, the subsequent documentation and software which manipulates the dataset is. The documentation of the dataset is made available with a CC-BY-4.0 license⁷ and the software under a MIT license⁸. The software itself contains two static methods of usage and reproduction to set up an environment to analyze this dataset. The dataset itself also requires all work to meet open science practices in addition to keeping a reference to the original dataset to avoid reproducibility issues on claims or malformed transformations.

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⁸ <https://opensource.org/licenses/MIT>

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